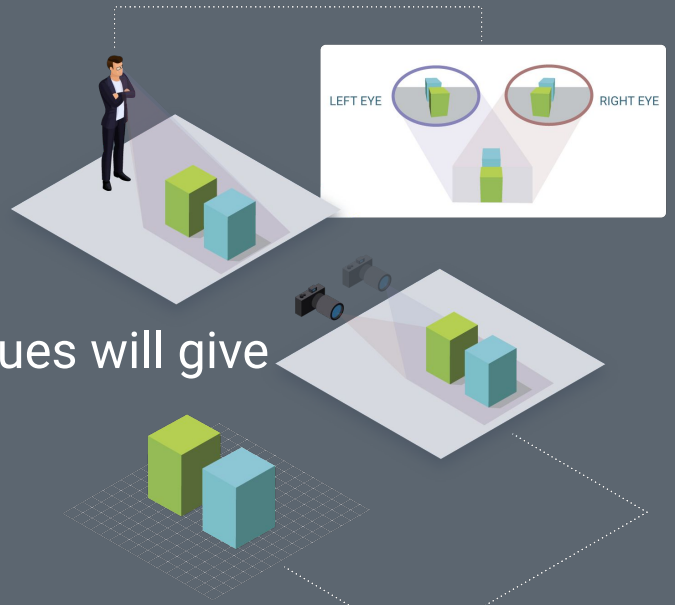
PIX4D

Measure From Images

Pix4D's Vision

"Evolution favours eyes that perceive the world in 3D"

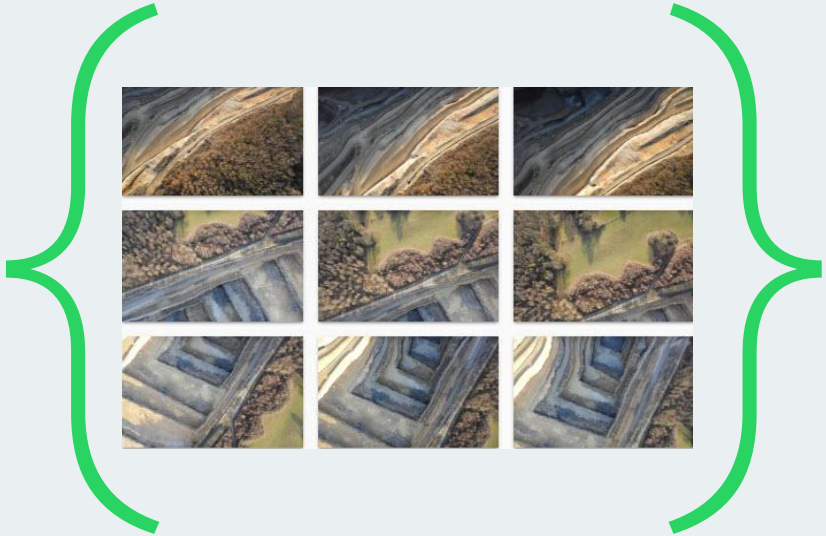
Christoph Strecha, Founder and CEO of Pix4D



At Pix4D, we believe that image-based techniques will give intelligent eyes to robotics of the future.

"44,354,881,622 Cameras in the world by 2022, at minimum", **LDV Capital**

Photogrammetry



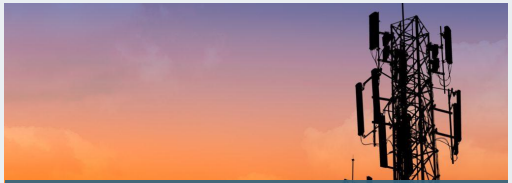


Applications


Surveying & Mapping


Construction


Mining & Aggregates


Industrial & Telecom inspection


Agriculture


Cultural heritage



Education

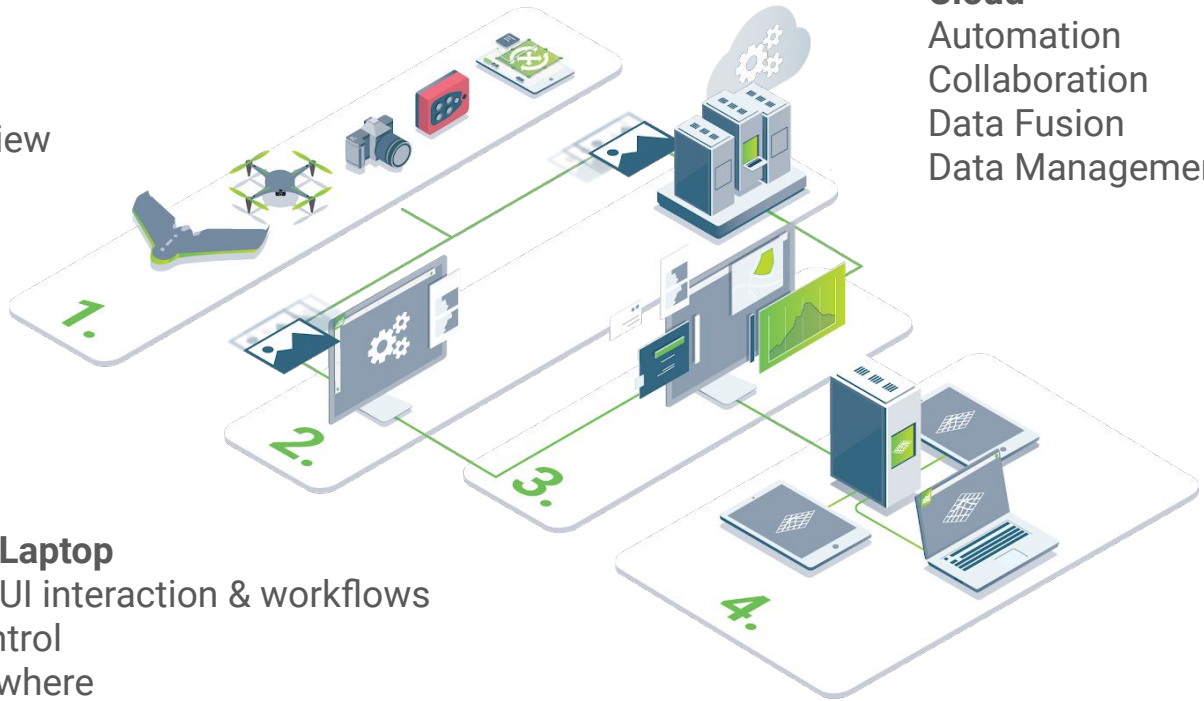

Public Safety


Humanitarian Aid & Disaster Response

Pix4D Platforms

Mobile
Field operations
Flight management
Data Capture & Review

Cloud
Automation
Collaboration
Data Fusion
Data Management & Storage



Desktop & Laptop
Complex GUI interaction & workflows
Quality Control
Works Anywhere

Pix4D at a Glance

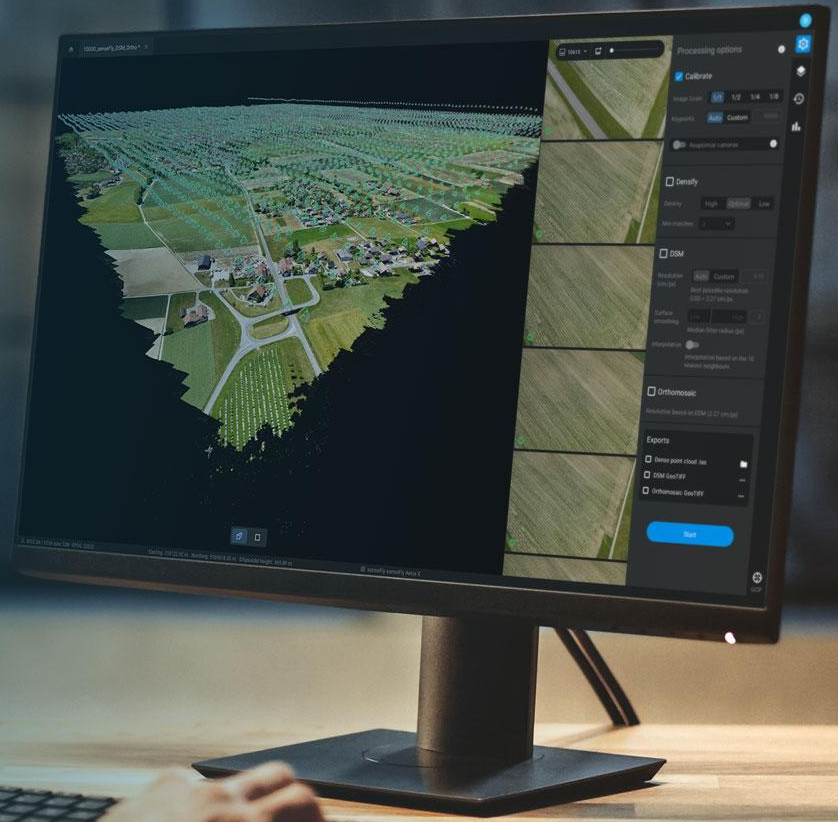


10 years of scientific research
2011 EPFL spin off
7 locations (3 R&D)
206 global employees
42 nationalities
26 business languages spoken
50k+ unique active users
12k+ weekly active users
100M+ acres modeled in
200+ countries

U.S.A. SWITZERLAND GERMANY CHINA SPAIN JAPAN

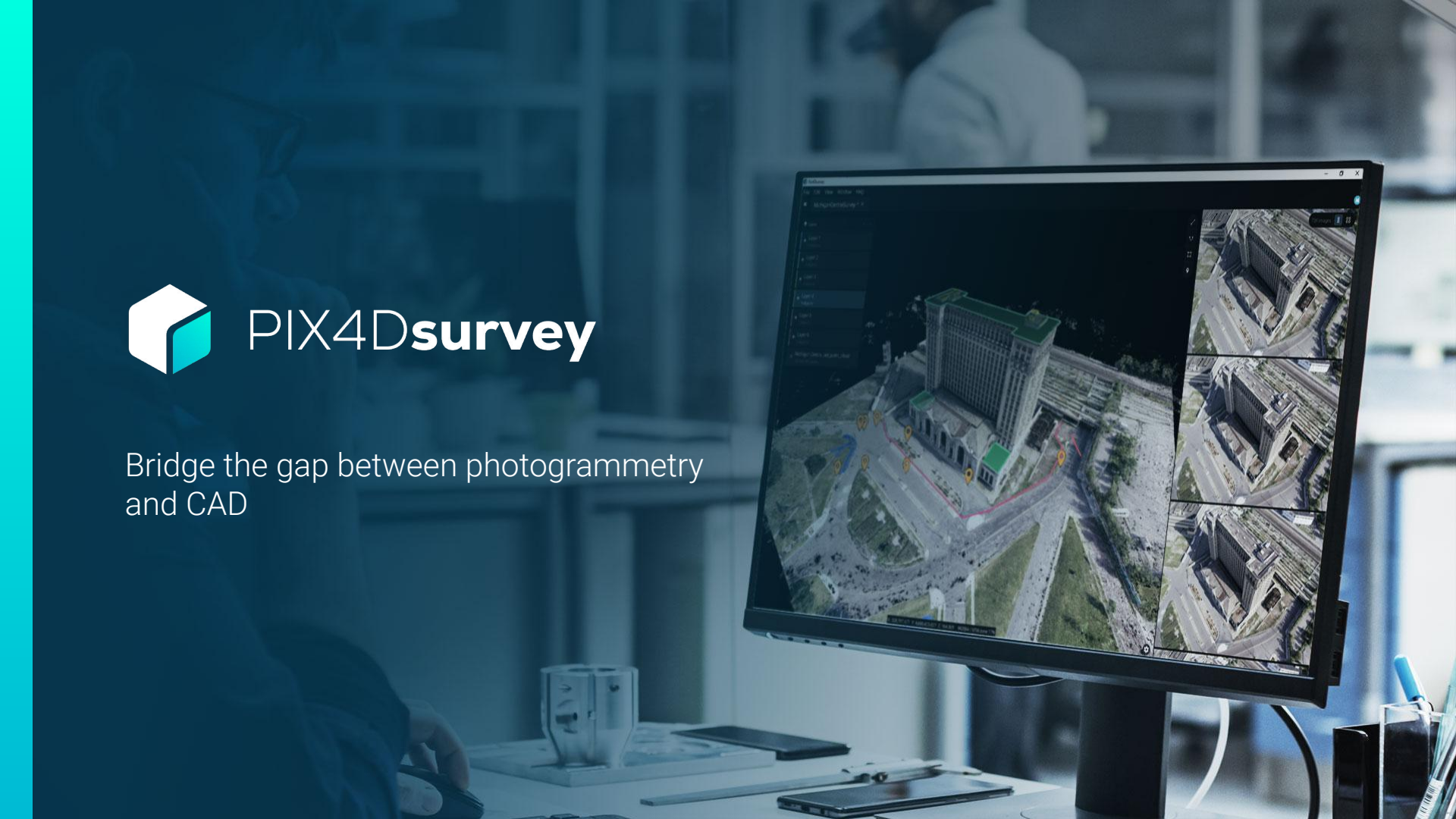
PIX4Dmatic

Next-generation photogrammetry software
for corridor and large scale mapping



PIX4Dsurvey

Bridge the gap between photogrammetry and CAD



PIX4Dfields

Advanced agriculture mapping software for aerial crop analysis and digital farming




PIX4Dinspect
Automate industrial inspection and asset management with drone images

PIX4Dreact

2D fast-mapping for emergency response and public safety



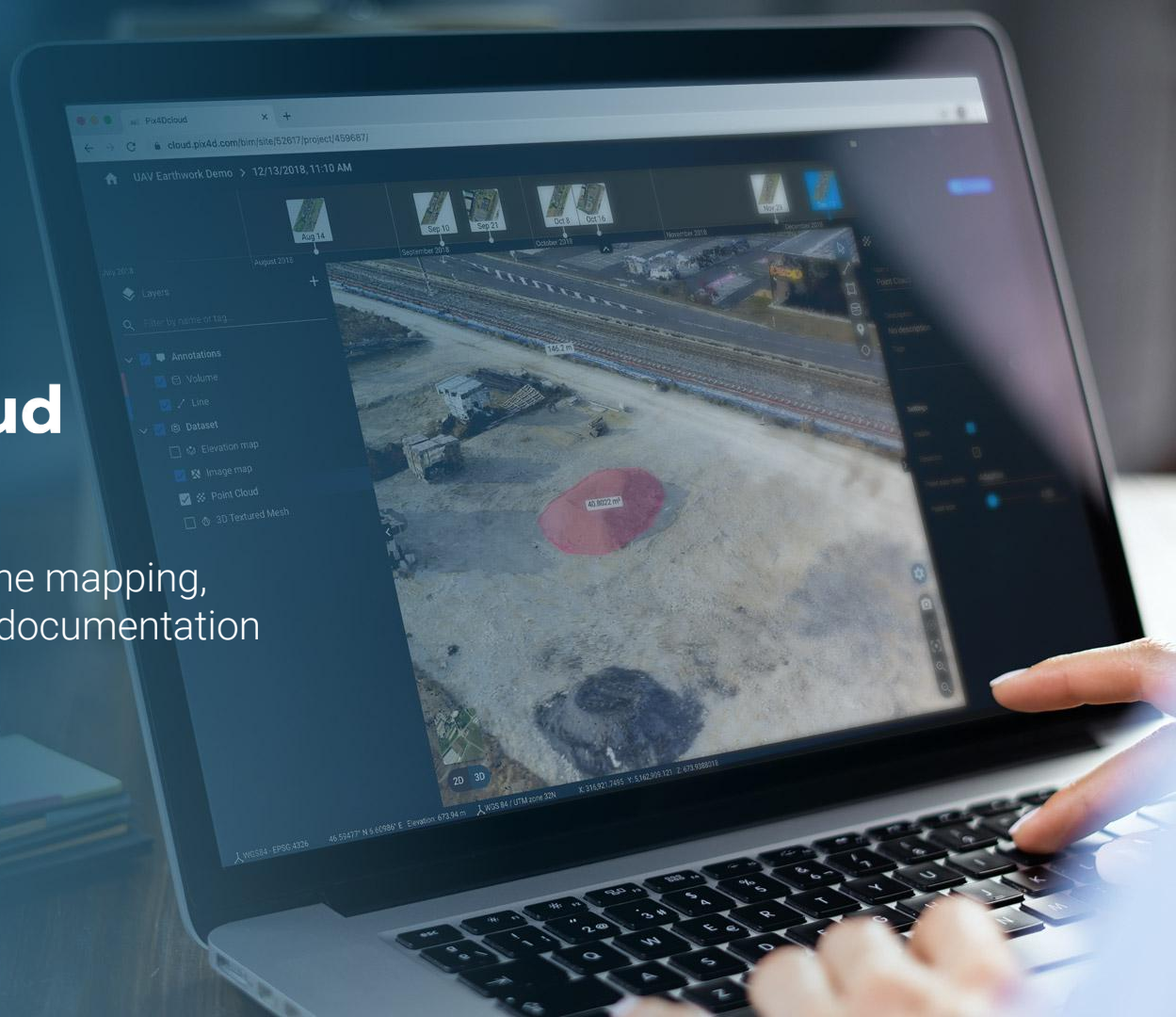
PIX4Dcatch

Turn your mobile device into a professional 3D scanner



PIX4Dcloud

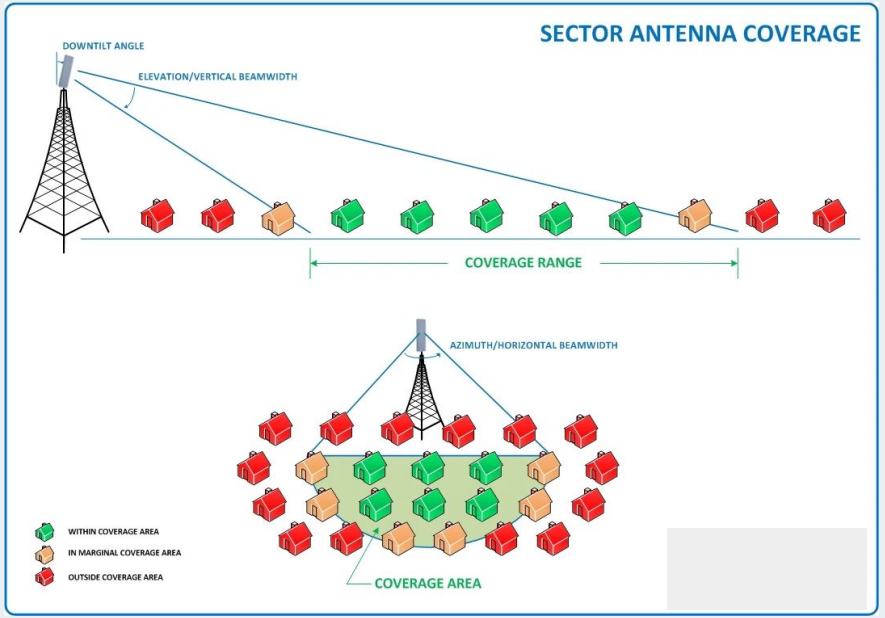
The online platform for drone mapping,
progress tracking, and site documentation



Measuring antenna pose

Why is this important?

- RF Engineers design positioning on tower to optimize coverage
- After installation the position is measured and compared to design requirements



Traditional workflow:

- Send technicians to the field
- Climb the tower (extremely dangerous)
- Use special hardware to measure pose
- Plan intervention if needed

Automation Opportunity: CV and ML

Automatic Equipment Measurements

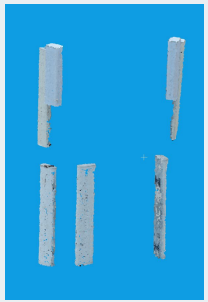




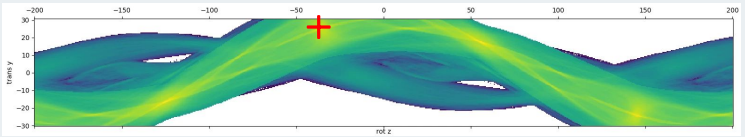
Step 1) Use state of the art ML to detect panel antennas in input images: segmentation problem

Step 2) The 2D segmentation masks are projected to 3D space, and the resulting 3D points are clustered and antenna detections refined

Step 3) 3D spatial symmetries are used to find high precision antenna poses



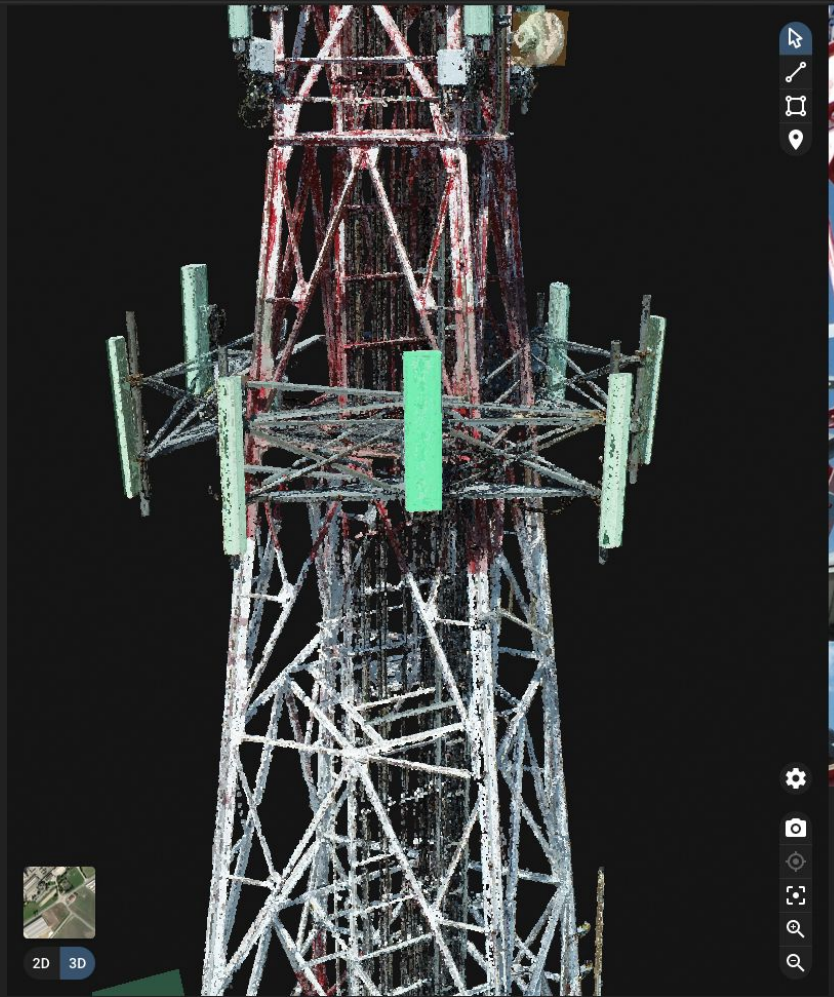
(Estimated pose best fit)



PIX4D

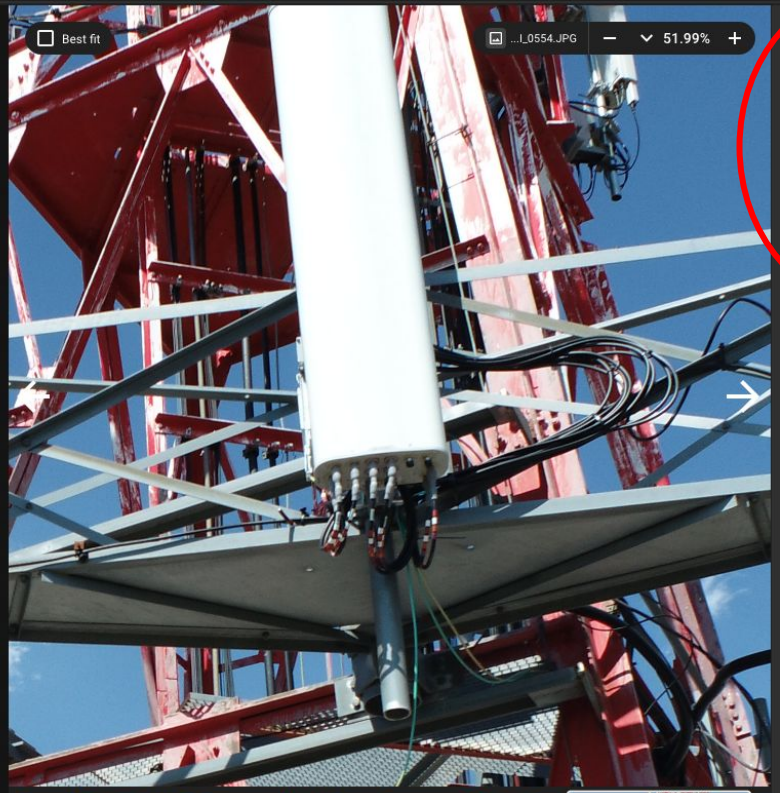
Pix4Dinspect

CT-005 6/7/2020 Completed All changes saved Share



2D 3D

Best fit ...I_0554.JPG 51.99%





Antenna Card

Automatic recognition data

Antenna Type	panel
Downtilt	1.55794°
Azimuth	234.039°
Plumb	0.76451°
Width	42.0 cm
Height	1.91 m
Depth	19.0 cm
Altitude (CL)	43.28 m

Model name

Kathrein [illegible]V02

ID

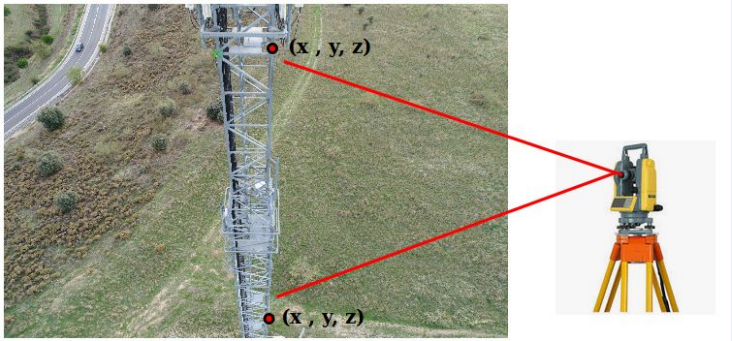
ANT15894

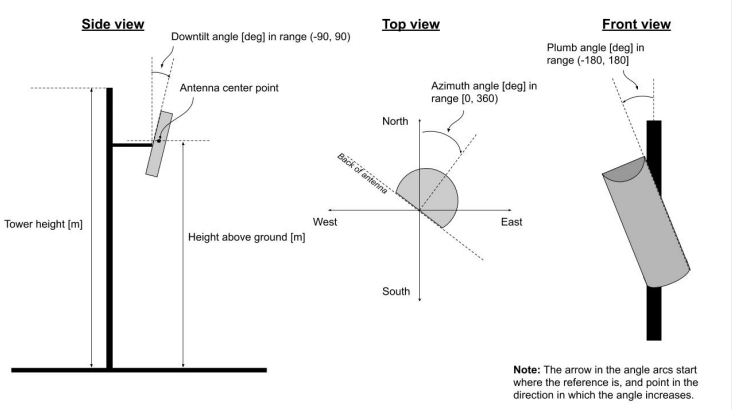
Frequencies

2300, 5200

Antenna is on

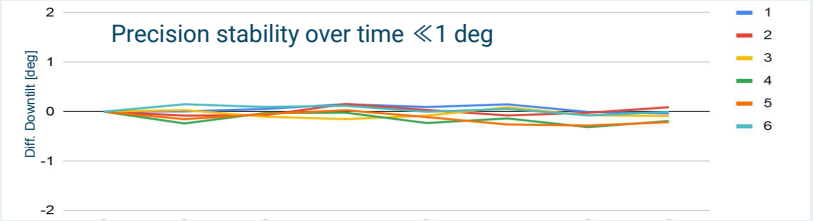
Results





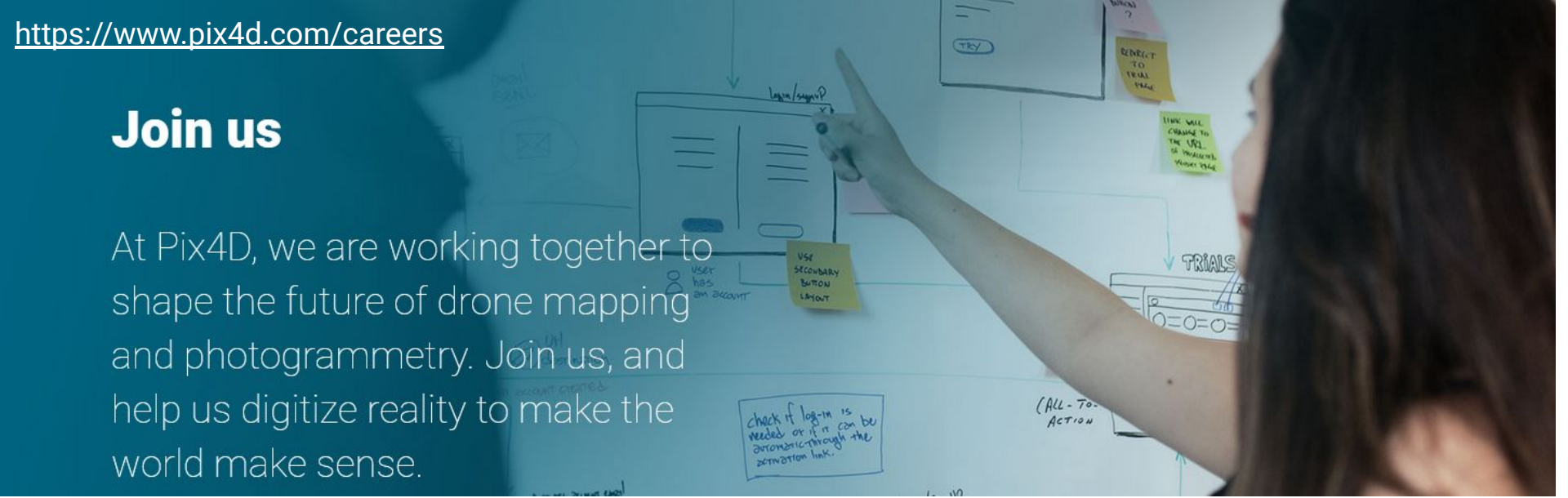
Data collected with total station: measure with extreme precision tower geometry

Analysis confirms that angles can be measured better than 1 deg for downtilt, better than 4 deg for azimuth



" [...] I'm happy to report the antenna bearings were about 1 degree different to the antenna arrangement drawing. [...] The antenna heights were within 10 cm of the drawing! Now that is impressive! [...]"
Early Beta tester Feedback

https://www.pix4d.com/careers



Join us

At Pix4D, we are working together to shape the future of drone mapping and photogrammetry. Join us, and help us digitize reality to make the world make sense.

Lausanne Office:
- R&D Engineer
- Backend Sw Engineer Intern
- Backend Sw Engineer
- Sw Engineer (C++, Python)
- Technical Lead Engineer
- DevOps Sw Engineer
- QA Assurance Engineer Mobile

Madrid Office:
- Technical Lead Engineer
- Backend Sw Engineer - RevOps

Berlin Office:
- Backend Sw Engineer
- Sr Backend Engineer
- Sw Engineer (C++/Qt/QML)
- Sw Architect (C++)